

Van-Thinh To

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RESEARCH PROFILE

Master's researcher in chemoinformatics developing machine-learning scoring functions, virtual-screening benchmarks, graph self-supervised learning, and generative models for small-molecule discovery. Experienced in large-scale PubChem/ChEMBL curation, molecular docking and reproducible Python workflows.

EDUCATION

Double Degree Master's Program-Erasmus Mundus Scholar

Université Paris Cité (France)

2025 - expected Jun 2027

Master of Bioinformatics

GPA: 18.05/20.00 (90%)

University of Milan (Italy)

2025 - expected Jun 2027

Master of Chemistry

GPA: 29.80/30.00 (99%)

University of Medicine and Pharmacy at Ho Chi Minh city (Vietnam)

2019 - 2024

Bachelor of Pharmacy (5-year program)

GPA: 3.31/4.00 (83%)

Rank top 4%

Thesis: A Knowledge-Guided Graph Self-Supervised Learning for Molecular Property Predictions

Grade: 10/10

Relevant training

Strasbourg Summer School in Cheminformatics (France)

June 2026

Vienna Summer School on Drug Design (Austria)

Sep 2023

RESEARCH EXPERIENCE

University Paris Cité (France)

Supervisor: Prof. Olivier Taboureau and Dr. Viet-Khoa Tran-Nguyen

Master's researcher

1. **Generative model coupling specific binding-affinity scoring with retrosynthesis planning** Jul 2026 - present

- Building specific binding-affinity scoring functions for optimizing generated molecules.
- Combining specific binding-affinity scoring with retrosynthesis planning.

2. **Large-scale PubChem/ChEMBL bioactivity curation and virtual screening benchmarking** Sep 2025 - present

- Automating large-scale retrieval of bioactivity data for *all* protein targets from PubChem and ChEMBL.
- Identifying dark chemical matter compounds for use as assumed inactives.
- Conducting virtual screening benchmarking studies across selected protein targets.

3. **Target-specific machine learning scoring functions for cystathionine β -synthase inhibitors** Jan 2026

- Curated data set from PubChem and ChEMBL.

University of Medicine and Pharmacy at Ho Chi Minh city (Vietnam)

Supervisor: Assoc. Prof. Tuyen Ngoc Truong and Dr. Tieu-Long Phan

Undergraduate researcher

1. **Generative AI for dual targets of EGFR and VEGFR** May 2024 - August 2025

- Developed a graph-based scoring function to evaluate generated molecules.

2. **A knowledge-guided graph for molecular representation** Dec 2023 - Jul 2024

- Proposed two knowledge vectors that incorporate orbital information for building a hierarchical molecular graph.
- Introduced pre-training tasks for self-supervised learning.
- Fine-tuned the model for molecular property prediction, including six classification and six regression data sets splitted by scaffold method.

3. **Application of molecular similarity, neural network and GNINA molecular docking in virtual screening of PD-L1 inhibitors** Jun 2023 - Dec 2023

- Proposed an artificial neural network model, and a molecular similarity assessment; utilized GNINA 1.0 molecular docking.
- Developed an automated pipeline for the molecular similarity model.
- Screened 15,235 DrugBank repurposing compounds and identified 22 hypothetical hit compounds.

SELECTED PUBLICATIONS

1. **To, V.-T.**, Van Nguyen, P.-C., Truong, G.-B., Phan, T.-M., Phan, T.-L., Fagerberg, R., Stadler, P. F. & Truong, T. N. KGG: Knowledge-Guided Graph Self-Supervised Learning to Enhance Molecular Property Predictions. *Journal of Chemical Information and Modeling*. doi:10.1021/acs.jcim.5c01068 (2025)
2. **To, V.-T.**, Phan, T.-L., Doan, B.-V. N., Nguyen, P.-C. V., Le, Q.-H. N., Nguyen, H.-H., Trinh, T.-C. & Truong, T. N. Innovative virtual screening of PD-L1 inhibitors: the synergy of molecular similarity, neural networks and GNINA docking. *Future Medicinal Chemistry*, 1–12. doi:10.1080/17568919.2024.2389773 (2024)
3. Nguyen, H.-H.[†], **To, V.-T.**[†], Truong, T. N. & Phan, T.-L. KITE: Knowledge-Infused Transformer Encoder for Molecular Property Prediction. *Preprint*. doi:10.26434/chemrxiv.15006882/v1 (2026)
4. Truong, C.-M., **To, V.-T.**, Janel, N., Dairou, J., Ballester, P. J., Taboureau, O. & Tran-Nguyen, V.-K. Enhancing virtual screening of cystathionine β -synthase inhibitors: benchmarking target-specific machine-learning scoring functions against state-of-the-art AI docking and co-folding approaches. *Journal of Cheminformatics*. doi:10.1186/s13321-026-01262-x (2026)
5. Van Nguyen, P.-C., **To, V.-T.**, Tran, N.-V. N., Phan, T.-L., Truong, T. N., Gärtner, T., Merkle, D. & Stadler, P. F. SynCat: molecule-level attention graph neural network for precise reaction classification. *Digital Discovery*. doi:10.1039/D5DD00367A (2026)
6. Phan, T.-M., Phan, T.-L., Van-Nguyen, P.-C., Le, H. S. L., **To, V.-T.**, Truong, T. N., Merkle, D. & Stadler, P. F. ProQSAR: A Modular and Reproducible Framework for Small-Data QSAR Modeling with Fit-and-Use Models. *Journal of Cheminformatics*. doi:10.1186/s13321-026-01175-9 (2026)

[†] Co-first authors; contributed equally. Full list on Google Scholar

TEACHING EXPERIENCE

University of Medicine and Pharmacy at Ho Chi Minh city (Vietnam)

Aug 2024 - Aug 2025

Teaching assistant and research mentoring - Department of Organic Chemistry, Faculty of Pharmacy

- Prepared training materials on computer-aided drug design for undergraduate students, such as RDKit, fingerprints, molecular descriptors, ANN, GNN.
- Mentored one undergraduate student on thesis project focused on self-supervised learning.
- Taught practical organic chemistry, with a focus on introduction of laboratory equipments, lab techniques and synthesis skills.

SKILLS

- **Cheminformatics:** RDKit; Machine Learning (scikit-learn); Deep learning (TensorFlow, PyTorch, GNN, ANN, CNN); Molecular Docking (GNINA); Virtual Screening; QSAR; Generative model (IRDiff); Property-matched decoy generation (DeepCoy); Databases (PDB, PubChem, ChEMBL); Visualization (PyMOL)
- **Programming skills:** Python, R, C, Linux/Unix command line, pytest, Git, Package management, deployment
- **English proficiency:** IELTS 7.0 overall (Academic Version) Nov 2024

GRANTS AND TALKS

- Oral presentation at 41st Scientific and Technical Conference of Pharmacy. 2025
- Poster presentation at Asian Federation for Pharmaceutical Sciences. 2023
- Grants (secretary): wrote research proposals for four university-funded projects, each awarded 1200 USD. 2025

SELECTED AWARDS

- **Pharmacy Scholarship for Elite Students** 2019 - 2024
Awarded a \$5000 scholarship over 5 years, equal to full tuition (acceptance rate: 10%)
- **Second prize in the nationwide Vietnamese Chemistry Olympiad** 2018, 2019
The biggest Chemistry competition for high school students in Vietnam, ranking 42/400+ overall
- **Odon Vallet Scholarship** 2019, 2024
Awarded to outstanding high school, university, and graduate students in Vietnam.
- **First prize** in the Chemistry Olympiad of Southern Vietnam for high school students, ranking 6/200+ overall 2017